KARUNGA JUNIOR SCHOOL
GRADE 9 MATHEMATICS
END OF TERM 1 EXAMINATION

Teacher-prepared assessment resource | Kenya CBC

Learner's Name:	_____	Grade:	9
Date:	_____	Time:	2 hours 30 minutes

INSTRUCTIONS

1. Answer ALL questions. Show all necessary working.

2. Mathematical tables and a scientific calculator may be used where appropriate, unless instructed otherwise.

3. Give answers to the degree of accuracy requested.

4. The examination consists of 10 questions and carries 100 marks.

SECTION A: SHORT-ANSWER QUESTIONS (40 MARKS)

1. Integers (8 marks)

(a) Evaluate: $-18 + 7 \times (-3) - (-5)$. (2 marks)

(b) A lift is on floor 4. It goes down 9 floors and then rises 6 floors. On which floor does it stop? (2 marks)

(c) Arrange the following integers in ascending order: −12, 5, −3, 0, 8. (2 marks)

(d) Find the difference between −17 and 9. (2 marks)

2. Cubes and Cube Roots (8 marks)

(a) Evaluate 6^3. (1 mark)

(b) Find $\sqrt[3]{729}$. (2 marks)

(c) Determine whether 1,331 is a perfect cube. Show your reasoning. (2 marks)

(d) A cube has a volume of 1,728 cm^3. Find its edge length. (3 marks)

3. Indices and Logarithms (8 marks)

(a) Simplify: $2^4 \times 2^3$. (2 marks)

(b) Evaluate: $5^0 + 3^2$. (2 marks)

(c) Express 1,000 as a power of 10. (1 mark)

(d) Evaluate $\log_{10} 100$. (1 mark)

(e) Simplify: $(a^5 \div a^2)$, where $a \neq 0$. (2 marks)

4. Compound Proportions and Rates of Work (8 marks)

(a) If 6 workers complete a task in 10 days, how many days would 12 workers take at the same rate? (3 marks)

(b) Four taps fill a tank in 6 hours. How long would 8 identical taps take, working together? (2 marks)

(c) A vehicle travels 180 km using 15 litres of fuel. How many litres are needed for 300 km at the same rate? (3 marks)

5. Matrices (8 marks)

Given A = [[2, 1], [−3, 4]] and B = [[1, 5], [2, −1]],

(a) Find A + B. (2 marks)

(b) Find 2A. (2 marks)

(c) Find A − B. (2 marks)

(d) State the order of matrix A. (2 marks)

SECTION B: STRUCTURED QUESTIONS (60 MARKS)

6. Integers in context (10 marks)

(a) At 6:00 a.m. the temperature was −4°C. By midday it had risen by 13°C and by midnight it had fallen by 8°C. Calculate the temperature at midnight. (4 marks)

(b) A trader's account showed a balance of KSh −2,450. The trader deposited KSh 5,000 and later withdrew KSh 1,875. Find the final balance. (3 marks)

(c) Explain why zero is neither positive nor negative. (3 marks)

7. Cubes, cube roots and indices (10 marks)

(a) Express 216 as a product of prime factors and hence determine its cube root. (4 marks)

(b) Simplify $(3^2 \times 3^4) \div 3^3$, giving your answer as a power and as an integer. (3 marks)

(c) A cubical container has an internal volume of 4,096 cm^3. Calculate the length of one internal edge. (3 marks)

8. Logarithms and applications (10 marks)

(a) Evaluate: $\log_{10} 1{,}000 + \log_{10} 100$. (3 marks)

(b) Given $\log_{10} 2 = 0.3010$, use logarithms to find $\log_{10} 20$. (3 marks)

(c) A population of bacteria doubles every hour. If there are initially 500 bacteria, write an expression for the population after 4 hours and calculate it. (4 marks)

9. Compound proportion and rates of work (10 marks)

(a) Eight machines produce 1,200 items in 5 hours. Assuming a constant rate, how many items will 12 machines produce in 8 hours? (5 marks)

(b) Fifteen workers can complete a project in 12 days. After working for 4 days, 5 workers leave. How many additional days will the remaining workers take to finish the work, assuming equal work rates? (5 marks)

10. Matrices and equation of a straight line (20 marks)

(a) Given P = [[1, 2], [3, 4]] and Q = [[2, 0], [−1, 5]], calculate PQ. (5 marks)

(b) A straight line passes through points A(2, 3) and B(6, 11). Find: (i) its gradient; (ii) its equation in the form $y = mx + c$. (6 marks)

(c) Determine the x-intercept and y-intercept of the line $2x + y = 8$. (4 marks)

(d) Complete a table of values for $y = 2x - 1$ when $x = -1, 0, 1, 2$. Plot the points on graph paper and draw the straight line. (5 marks)

MARKING SCHEME
GRADE 9 MATHEMATICS END OF TERM 1

TOTAL: 100 MARKS

1. Integers (8 marks)

(a) $-18 + 7(-3) - (-5) = -18 - 21 + 5 = -34$. (2)

(b) $4 - 9 + 6 = 1$; floor 1. (2)

(c) $-12, -3, 0, 5, 8$. (2)

(d) $9 - (-17) = 26$. (2)

2. Cubes and Cube Roots (8 marks)

(a) 216. (1)

(b) $\sqrt[3]{729} = 9$. (2)

(c) $1{,}331 = 11 \times 11 \times 11 = 11^3$; therefore it is a perfect cube. (2)

(d) Edge = $\sqrt[3]{1{,}728} = 12$ cm. (3)

3. Indices and Logarithms (8 marks)

(a) $2^4 \times 2^3 = 2^7 = 128$. (2)

(b) $1 + 9 = 10$. (2)

(c) 10^3. (1)

(d) 2. (1)

(e) $a^5 \div a^2 = a^3$. (2)

4. Compound Proportions and Rates of Work (8 marks)

(a) Inverse proportion: $6 \times 10 = 12 \times d$; $d = 5$ days. (3)

(b) $4 \times 6 = 8 \times t$; $t = 3$ hours. (2)

(c) 180 km uses 15 L; rate = 15/180 L per km. For 300 km: $300 \times 15/180 = 25$ L. (3)

5. Matrices (8 marks)

(a) [[3, 6], [−1, 3]]. (2)

(b) [[4, 2], [−6, 8]]. (2)

(c) [[1, −4], [−5, 5]]. (2)

(d) 2×2. (2)

6. Integers in context (10 marks)

(a) $-4 + 13 - 8 = 1$°C. Award method (2), correct result and unit (2).

(b) −2,450 + 5,000 − 1,875 = KSh 675. Award correct operations (2), final balance (1).

(c) Zero is the reference point separating positive and negative numbers; it has no positive or negative sign and is not greater or less than itself. Award up to 3 for clear explanation.

7. Cubes, cube roots and indices (10 marks)

(a) $216 = 2^3 \times 3^3 = (2 \times 3)^3$; cube root = 6. Factorisation (2), grouping (1), answer (1).

(b) $(3^2 \times 3^4) \div 3^3 = 3^{(2+4-3)} = 3^3 = 27$. (3)

(c) $\sqrt[3]{4{,}096} = 16$ cm because $16^3 = 4{,}096$. (3)

8. Logarithms and applications (10 marks)

(a) 3 + 2 = 5. (3)

(b) $\log_{10} 20 = \log_{10}(2 \times 10) = \log_{10}2 + \log_{10}10 = 0.3010 + 1 = 1.3010$. (3)

(c) $P = 500 \times 2^4 = 500 \times 16 = 8{,}000$ bacteria. Expression (2), calculation (2).

9. Compound proportion and rates of work (10 marks)

(a) Output is directly proportional to machines and hours: $1{,}200 \times (12/8) \times (8/5) = 2{,}880$ items. Correct proportional method (3), answer with unit (2).

(b) Total work = 15 × 12 = 180 worker-days. Work done in 4 days = 15 × 4 = 60 worker-days; remaining = 120. Remaining workers = 10; days = 120/10 = 12 days. (5)

10. Matrices and equation of a straight line (20 marks)

(a) PQ = [[1×2+2×(−1), 1×0+2×5], [3×2+4×(−1), 3×0+4×5]] = [[0, 10], [2, 20]]. (5)

(b) Gradient $m = (11-3)/(6-2) = 8/4 = 2$. Using $y = 2x + c$ and (2,3): $3 = 4 + c$, $c = -1$. Equation $y = 2x - 1$. Gradient (3), equation (3).

(c) x-intercept: y=0, 2x=8, x=4 → (4,0). y-intercept: x=0, y=8 → (0,8). (4)

(d) Values: x = −1, 0, 1, 2; y = −3, −1, 1, 3. Table/values (2), correctly plotted points (2), straight line (1).

GENERAL MARKING GUIDANCE

Award method marks where a correct mathematical process is shown, even if an arithmetic error affects the final answer.

Accept equivalent correct methods and notation.

Award units where required. Do not award full accuracy marks when a required unit is omitted.

For graphs, use a consistent scale, labelled axes and accurately plotted points; award marks according to the breakdown in Question 10(d).

Teachers should verify that all assessed sub-strands have been covered before administering the examination. This paper follows the Term 1 sequence in the teacher-prepared Grade 9 Mathematics scheme and is not an official KICD examination.

END OF EXAMINATION